5. Related work / 1 of 3

Five recent EEG/GNN papers; all numbers below are author-reported, not reproduced project results.

R1 | Self-Supervised Learning for Anomalous Channel Detection in EEG Graphs: Application to Seizure Analysis

Thi Kieu Khanh Ho; Narges Armanfard | 2023 (AAAI; preprint 2022)

Dataset / area	TUSZ v1.5.2, 19 channels; Seizure clip and channel detection
Method / GNN	EEG-CGS: GCN autoencoder, contrastive and reconstruction losses
Graph	Electrodes are nodes; distance, random, correlation and directed-transfer graph variants.
Evaluation	Normal-only training: 35,019 clips; held-out test: 8,848 clips including seizures.
Reported metrics	Combined graphs: clip F1 0.534, sensitivity 0.501, specificity 0.974 (Table 3).
Strength	Learns from normal data and compares graph variants.
Limitation	Channel-localization task requires multichannel signals; cannot transfer unchanged to a single channel.
Gap / relation	Contribution of temporal edges in single-channel graphs is not answered by this multichannel experiment. Motivates controlled edge-rule ablations; numbers are not Bonn targets.
Evidence	Full text, graph construction and Tables 1/3.

R2 | Dynamic GNNs for Precise Seizure Detection and Classification from EEG Data

Arash Hajisafi; Haowen Lin; Yao-Yi Chiang; Cyrus Shahabi | 2024 (arXiv version reviewed)

Dataset / area	TUSZ v1.5.2; Seizure detection and type classification
Method / GNN	NeuroGNN: modified residual GCN, BiGRU and hierarchical pooling
Graph	Electrodes and brain-region meta-nodes; temporal, spatial and semantic edges.
Evaluation	Separate test set; training portion divided 9:1 for training/validation; three random runs.
Reported metrics	With pretraining: detection AUROC 0.876 +/- 0.002; classification weighted F1 0.792 +/- 0.004 (Table 2).
Strength	Tests each graph component through ablation.
Limitation	Requires electrode geometry and brain-region information absent from isolated single-channel segments.
Gap / relation	Does temporal order alone add value when spatial information is unavailable? Supports matching feature inputs and changing one edge mechanism at a time.
Evidence	Full text, Sections 3-4 and Table 2.

5. Related work / 2 of 3

R3 | Sequential graph convolutional network and DeepRNN based hybrid framework for epileptic seizure detection from EEG signal

Ferdaus Anam Jibon; A. R. Jamil Chowdhury; Mahadi Hasan Miraz; Hwang Ha Jin; Mayeen Uddin Khandaker; Sajia Sultana; Sifat Nur; Fazlul Hasan Siddiqui; AHM Kamal; Mohammad Salman; Ahmed A. F. Youssef | 2024

Dataset / area	CHB-MIT and TUH; Seizure detection
Method / GNN	Sequential GCN plus DeepRNN
Graph	Pearson-correlation graph; periodogram-based signal processing.
Evaluation	CHB-MIT patient-by-patient experiments: 5-fold CV, repeated 10 times. Subject-disjoint testing not established by this description.
Reported metrics	Table 2 accuracy: CHB-MIT 99.007%; TUH 98.08%.
Strength	Compares the hybrid against its individual components.
Limitation	Reporting inconsistencies: TUH sensitivity/specificity entries differ between Tables 2 and 5.
Gap / relation	Need auditable splits and consistent metric reporting before comparing architectures. Use reproducible split manifests and distinguish patient-specific from unseen-patient claims.
Evidence	Full text, Algorithm 1 and Tables 2/3/5. Results transcribed, not independently reproduced.

R4 | Multi-branch fusion graph neural network based on multi-head attention for childhood seizure detection

Yang Li; Yang Yang; Shangling Song; Hongjun Wang; Mengzhou Sun; Xiaoyun Liang; Penghui Zhao; Baiyang Wang; Na Wang; Qiyue Sun; Zijuan Han | 2024

Dataset / area	CHB-MIT; Childhood seizure detection
Method / GNN	MGCNA: multi-branch GCN with multi-head attention
Graph	Spatial-distance, functional-correlation and trainable graphs over channels.
Evaluation	Patient-specific 5-fold CV; patient-independent leave-one-subject-out evaluation.
Reported metrics	Table 1: patient-specific accuracy 99.32%, F1 97.76%. Patient-independent accuracy conflicts: 89.45% in Table 2 versus 85.45% in Table 4.
Strength	Separately studies patient-specific and patient-independent performance.
Limitation	Requires channel locations; reported cross-patient accuracy is internally inconsistent.
Gap / relation	Need simple graph ablations with explicit generalization limits. Supports reporting source grouping and computational cost rather than chasing a published accuracy.
Evidence	Full text, Sections 2.8/4.3 and Tables 1/2/4.

5. Related work / 3 of 3

R5 | Lightweight Graph Neural Network Framework for Seizure Detection Using Single-Channel EEG

Shobha Sharma; Tapan Kumar Gandhi | 2026 (online 17 April; PReMI 2025 proceedings)

Dataset / area	Bonn, five categories Z/N/F/S/O; EEG segment classification
Method / GNN	Lightweight GCN
Graph	Eight non-overlapping temporal segments per recording; feature-similarity edges.
Evaluation	Ten random seeds; split proportions and subject grouping not disclosed in the accessible abstract.
Reported metrics	Across seeds: accuracy 0.87 +/- 0.0147; macro-F1 0.8681 +/- 0.0157. Best-seed accuracy 91%.
Strength	Direct precedent for a single-channel temporal-segment graph.
Limitation	Only publisher abstract accessible; exact features, edges and split cannot be verified.
Gap / relation	The accessible evidence does not isolate the value of temporal-adjacency edges versus similarity edges. Closest prior work. Propose a controlled extension, not a claim that Bonn GNNs are new.
Evidence	Publisher abstract only; full chapter is subscription content.

6. Research gap and testable question

The reviewed multichannel GNNs use electrode relationships that cannot be copied directly to these single-channel segments. R5 already constructs a Bonn graph from eight temporal parts, so applying a GNN to Bonn is not a new contribution. The accessible evidence does not isolate temporal adjacency from feature-similarity edges; this motivates a controlled project experiment, not a claim of proven global novelty.

Question	On a fixed, documented segment-level Bonn partition, does adding temporal-neighbour edges to similarity edges improve macro-F1 over temporal-only and similarity-only GCNs under identical features and training budget?
Proposed graph	One released EEG segment per graph; eight contiguous parts as nodes. numpy.array_split preserves 4,097 values: 513 + seven parts of 512.
Node / edge design	19 descriptors per node; temporal chain, feature-similarity k-nearest neighbours, or their union. Scale features using training nodes only; select k on validation data.
Controlled comparison	Same splits, node features, GCN size, optimization budget and seeds. Also compare Logistic Regression, SVM and Random Forest on segment features.
Outcome	Macro-F1, class recall, accuracy, time and parameter count; a negative result is valid.

Temporal edges encode order, not anatomical brain connectivity. Keep every edge within its original segment and keep all derived parts together. Subject-level generalization remains unverified because patient/parent-recording metadata are absent. The instructor must decide whether this limitation and temporal graph satisfy Track 1 before later modelling.